SF_6 Global Plasma Model

Complete Mathematical and Chemical Documentation

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1 Model Overview

This is a zero-dimensional (global, volume-averaged) model for an SF_6 inductively coupled plasma (ICP) discharge. The model solves **18 coupled ordinary differential equations** for:

- 13 gas-phase species densities (6 neutrals, 4 positive ions, 2 negative ions, electrons)
- 1 electron temperature
- 4 surface coverages

The system is time-integrated from pre-discharge initial conditions to steady state using the BDF (Backward Differentiation Formula) method.

2 Species

2.1 Gas-Phase Species

Index	Species	Type	Mass (amu)
0	SF_6	Parent gas (neutral)	146
1	SF_5	Neutral radical	127
2	SF_4	Neutral radical	108
3	SF_3	Neutral radical	89
4	F_2	Neutral molecule	38
5	F	Neutral atom	19
6	SF_5^+	Positive ion	127
7	SF_4^+	Positive ion	108
8	SF_3^+	Positive ion	89
9	F_2^+	Positive ion	38
10	SF_6^-	Negative ion	146
11	F^-	Negative ion	19
12	e^-	Electron	5.486×10^{-4}

Table 1: Gas-phase species included in the model.

2.2 Surface Species

The reactor walls are described by four coverage fractions:

$$\theta_{\mathrm{F}}, \quad \theta_{\mathrm{SF3}}, \quad \theta_{\mathrm{SF4}}, \quad \theta_{\mathrm{SF5}}, \quad \theta_{\mathrm{bare}} = 1 - \theta_{\mathrm{F}} - \theta_{\mathrm{SF_3}} - \theta_{\mathrm{SF_4}} - \theta_{\mathrm{SF_5}} \tag{1}$$

3 Reactor Geometry

The ICP tube (radius 19 cm, height 21 cm) and diffusion chamber (radius 19 cm, height 17 cm) are modeled as a single equivalent cylinder:

$$R = 0.19\,\mathrm{m}, \qquad L = 0.38\,\mathrm{m} \tag{2}$$

$$V = \pi R^2 L = 0.04309\,\mathrm{m}^3 \tag{3}$$

$$A_{\mathrm{ax}} = 2\pi R^2 = 0.2268\,\mathrm{m}^2 \quad \text{(top + bottom)} \tag{4}$$

$$A_{\mathrm{rad}} = 2\pi R L = 0.4536\,\mathrm{m}^2 \quad \text{(side wall)} \tag{5}$$

$$A = A_{\mathrm{ax}} + A_{\mathrm{rad}} = 0.6804\,\mathrm{m}^2 \tag{6}$$

The effective diffusion length is:

$$\Lambda = \left[\left(\frac{\pi}{L}\right)^2 + \left(\frac{2.405}{R}\right)^2\right]^{-1/2} = 0.0661\,\mathrm{m} \tag{7}$$

4 Gas-Phase Reactions

4.1 Electron-Impact Rate Coefficient Formula

All electron-impact rate coefficients use the parametric form with Druyvesteyn EEDF parameters:

$$\boxed{k(T_e) = \exp\left(A + B\ln T_e + \frac{C}{T_e} + \frac{D}{T_e^2} + \frac{E}{T_e^3}\right) \quad [\mathrm{m^3\,s^{-1}}]} \tag{8}$$

where T_e is the electron temperature in eV.

4.2 Neutral Dissociations (G1–G7)

These reactions break SF_6 and its fragments into smaller species by electron impact.

	Reaction	$\varepsilon_{\mathrm{th}}$ (eV)	A	B	C	D	E
G1	$SF_6 + e \to SF_5 + F + e$	9.6	−29.35	−0.238	−14.11	−15.25	−1.204
G2	$SF_6 + e \to SF_4 + 2F + e$	12.1	−31.61	−0.259	−10.00	−31.24	−0.713
G3	$SF_6 + e \to SF_3 + 3F + e$	16.0	−40.26	3.135	5.895	−64.68	0.261
G4	$SF_5 + e \to SF_4 + F + e$	9.6	Same as G1				
G5	$SF_4 + e \to SF_3 + F + e$	9.6	Same as G1				
G6	$F_2 + e \to 2F + e$	3.16	−31.44	−0.699	−5.170	−1.389	−0.065
G7	$F_2 + e \to 2F + e$	4.34	−33.44	−0.276	−3.564	−3.946	−0.039

4.3 Ionizations (G8–G16)

These reactions produce positive ions and free electrons.

	Reaction	$\varepsilon_{\mathrm{th}}$	A	B	C	D	E
G8	$SF_6 + e \to SF_5^+ + F + 2e$	15.5	−33.66	1.212	−4.594	−56.66	−0.323
G9	$SF_6 + e \to SF_4^+ + 2F + 2e$	18.5	−37.14	1.515	−4.829	−80.42	−0.792
G10	$SF_6 + e \to SF_3^+ + 3F + 2e$	20.0	−36.82	1.740	−0.105	−98.18	0.106
G11	$SF_5 + e \to SF_5^+ + 2e$	11.2	−34.92	1.487	−2.377	−29.71	−0.145
G12	$SF_5 + e \to SF_4^+ + F + 2e$	14.5	−36.27	1.892	−1.387	−50.87	−0.076
G13	$SF_4 + e \to SF_4^+ + 2e$	13.0	−32.95	0.876	−10.19	−31.21	−3.989
G14	$SF_4 + e \to SF_3^+ + F + 2e$	14.5	−32.75	0.822	−10.82	−40.59	−4.274
G15	$SF_3 + e \to SF_3^+ + 2e$	11.0	−35.55	1.750	−2.086	−28.70	−0.136
G16	$F_2 + e \to F_2^+ + 2e$	15.69	−35.60	1.467	−6.140	−57.14	−0.486

4.4 Attachments (G17–G19)

These reactions produce negative ions by capturing an electron.

	Reaction	$\varepsilon_{\mathrm{th}}$	A	B	C	D	E
G17	$SF_6 + e \to SF_5 + F^-$	0.1	−33.43	−1.173	−0.561	0.180	−0.015
G18	$SF_6 + e \to SF_6^-$	0.0	−33.46	−1.500	0.000	−0.002	0.0
G19	$F_2 + e \to F + F^-$	0.0	−33.31	−1.487	−0.280	0.011	0.0

4.5 Detachments, Ion–Ion Recombination, Ion–Molecule (G20–G50)

Detachments (G20–G21): Negative ions release their electron upon collision with any neutral N:

$$\text{G20:}\quad F^- + N \to F + N + e^- \qquad k = e^{-44.39} = 4.77\times10^{-20}\,\mathrm{m^3/s} \tag{9}$$

$$\text{G21:}\quad SF_6^- + N \to SF_6 + N + e^- \qquad k = e^{-44.98} = 2.58\times10^{-20}\,\mathrm{m^3/s} \tag{10}$$

Ion–ion recombination (G42–G49): All positive–negative ion pairs recombine at the same rate:

$$I^+ + J^- \to I + J, \qquad k_{ii} = e^{-29.93} = 1.10 \times 10^{-13}\,\mathrm{m^3/s} \tag{11}$$

Ion–molecule reaction (G50):

$$SF_6 + SF_5^+ \to SF_6 + SF_3^+ + F_2, \qquad k = e^{-39.65} = 5.42 \times 10^{-18}\,\mathrm{m^3/s} \tag{12}$$

4.6 Neutral Recombinations — Troe Fall-Off Formula (G35–G41)

Reactions G35–G37 use the **pressure-dependent Troe fall-off formula** from Ryan & Plumb (1990):

$$\boxed{k = \frac{k_0[M]}{1 + k_0[M]/k_\infty} \cdot F_c^{\left(1+\left[\log_{10}\left(\frac{k_0[M]}{k_\infty}\right)\right]^2\right)^{-1}}} \tag{13}$$

where $[M]$ is the total gas number density (in $\mathrm{cm^{-3}}$). The parameters for SF_6 as third body are:

	Reaction	k_0 ($\mathrm{cm^6/s}$)	k_∞ ($\mathrm{cm^3/s}$)	F_c	Regime at ~1 Pa
G35	$F + SF_5 \to SF_6$	3.4×10^{-23}	1×10^{-11}	0.43	High-pressure limit
G36	$F + SF_4 \to SF_5$	3.7×10^{-28}	5×10^{-12}	0.46	Low-pressure limit
G37	$F + SF_3 \to SF_4$	2.8×10^{-26}	2×10^{-11}	0.47	Fall-off region

Table 5: Troe fall-off parameters from Ryan & Plumb (1990), Table III.

Remaining recombinations use constant rate coefficients:

$$\text{G38–G40:} \quad F_2 + SF_x \to SF_{x+1} + F \qquad k = e^{-46.41} = 6.82 \times 10^{-21}\,\mathrm{m^3/s} \tag{14}$$

$$\text{G41:} \quad SF_5 + SF_5 \to SF_6 + SF_4 \qquad k = e^{-41.50} = 9.48 \times 10^{-19}\,\mathrm{m^3/s} \tag{15}$$

5 Surface Reactions

5.1 Adsorption (S1–S4)

	Reaction	Probability
S1	F + bare → F(s)	0.150
S2	SF_3 + bare → SF_3(s)	0.080
S3	SF_4 + bare → SF_4(s)	0.080
S4	SF_5 + bare → SF_5(s)	0.080

5.2 Fluorination (S5–S7)

Gas-phase F reacts with adsorbed SF_x. Only SF_6 (fully saturated) returns to gas.

	Reaction	Probability
S5	F + SF_3(s) → SF_4(s)	0.500
S6	F + SF_4(s) → SF_5(s)	0.200
S7	F + SF_5(s) → SF_6(gas)	0.025

5.3 Surface Recombinations (S8–S11)

Gas-phase species recombine with adsorbed F.

	Reaction	Probability
S8	$F + F(s) \rightarrow F_2(gas)$	0.500
S9	$SF_3 + F(s) \rightarrow SF_4(gas)$	1.000
S10	$SF_4 + F(s) \rightarrow SF_5(gas)$	1.000
S11	$SF_5 + F(s) \rightarrow SF_6(gas)$	1.000

5.4 Ion–Surface Interactions (S12–S31)

Positive ions arrive at the wall with the Bohm flux and interact depending on surface coverage:

$$\text{On bare:} \quad SF_x^+ + \text{bare} \rightarrow SF_x(s) \tag{16}$$

$$\text{On F(s):} \quad SF_x^+ + F(s) \rightarrow SF_x(s) + F(gas) \tag{17}$$

$$\text{On } SF_y(s)\text{:} \quad SF_x^+ + SF_y(s) \rightarrow SF_x(s) + SF_y(gas) \tag{18}$$

Branching is proportional to θ_{bare}, θ_{F}, and θ_{SF_y}.

5.5 Deposition (S32–S40)

Gas-phase SF_x deposits on sites covered by SF_y, forming a permanent fluoro-sulfur film:

$$SF_x + SF_y(s) \rightarrow SF_x(s) + P(s), \qquad s = 0.030 \tag{19}$$

for $x, y \in \{3, 4, 5\}$ (9 reactions total).

6 Governing Equations

6.1 Neutral Species Mass Balance

For each neutral species i:

$$\boxed{\frac{dn_i}{dt} = \frac{Q_{f,i}}{V} - k_{\text{pump}}\, n_i + \sum_j R_{i,j}^{\text{gas}} + \sum_k R_{i,k}^{\text{surf}}} \tag{20}$$

where $Q_{f,i}/V$ is the volumetric feed rate (SF_6 only), and:

$$k_{\text{pump}} = \frac{Q_{\text{total}}}{n_0 \cdot V}, \qquad n_0 = \frac{p_{\text{OFF}}}{k_B T_{\text{gas}}} \tag{21}$$

Example — SF_6 balance:

$$\begin{aligned} \frac{dn_{SF_6}}{dt} = \frac{Q_{SF_6}}{V} &- k_{\text{pump}}\, n_{SF_6} \\ &+ k_{G35}\, n_F\, n_{SF_5} + k_{G38}\, n_{F_2}\, n_{SF_5} + k_{G41}\, n_{SF_5}^2 \\ &+ R_{S7} + R_{S11} + k_{\text{det}}\, n_{SF_6^-}\, n_N + k_{ii}\, n_{SF_6^-}\, n_+ \\ &- (k_1 + k_2 + k_3 + k_8 + k_9 + k_{10} + k_{17} + k_{18})\, n_e\, n_{SF_6} \end{aligned} \tag{22}$$

6.2 Positive Ion Mass Balance

$$\frac{dn_{i^+}}{dt} = \sum(\text{ionization}) - \nu_{\text{wall}}\, n_{i^+} - k_{ii}\, n_{i^+}\, n_- - k_{\text{pump}}\, n_{i^+} \tag{23}$$

6.3 Negative Ion Mass Balance

Negative ions do **not** reach the walls (confined by sheath potential) and are **not** pumped:

$$\frac{dn_{j^-}}{dt} = \sum(\text{attachment}) - k_{\text{det}}\, n_{j^-}\, n_N - k_{ii}\, n_{j^-}\, n_+ \tag{24}$$

6.4 Charge Neutrality

$$\boxed{n_e = \sum_i n_{i^+} - \sum_j n_{j^-}}, \qquad \frac{dn_e}{dt} = \sum_i \frac{dn_{i^+}}{dt} - \sum_j \frac{dn_{j^-}}{dt} \tag{25}$$

6.5 Electron Energy (Power) Balance

$$\boxed{\frac{d}{dt}\left(\tfrac{3}{2} n_e T_e\right) = \frac{\eta\, P_{\text{abs}}}{V\, e} - n_e\, \mathcal{P}_{\text{coll}} - \mathcal{P}_{\text{wall}} - \mathcal{P}_{\text{pump}}} \tag{26}$$

Using the product rule, this gives an ODE for T_e:

$$\frac{dT_e}{dt} = \frac{2}{3n_e}\left[\frac{\eta\, P_{\text{abs}}}{Ve} - n_e\, \mathcal{P}_{\text{coll}} - \mathcal{P}_{\text{wall}} - \mathcal{P}_{\text{pump}}\right] - \frac{T_e}{n_e}\frac{dn_e}{dt} \tag{27}$$

Power coupling efficiency: $\eta = 0.50$.
Collisional losses (per electron, in $\text{eV}\,\text{s}^{-1}$):

$$\mathcal{P}_{\text{coll}} = \sum_j k_j(T_e)\, n_{\text{target},j}\, \varepsilon_j \tag{28}$$

where $\varepsilon_j = \varepsilon_{\text{threshold}}$ for inelastic reactions and $\varepsilon_j = 3(m_e/M_n)T_e$ for elastic collisions.
Wall losses ($\text{eV}\,\text{m}^{-3}\,\text{s}^{-1}$):

$$\mathcal{P}_{\text{wall}} = \sum_{i^+} \Gamma_{i^+}^{\text{wall}} \left(\tfrac{1}{2}T_e + V_{s,i}\right) + \Gamma_{\text{ion}}^{\text{total}} \cdot 2T_e \tag{29}$$

where the sheath voltage for each ion species is:

$$V_{s,i} = \frac{T_e}{2} \ln\left(\frac{M_i}{2\pi m_e}\right) \tag{30}$$

Pumping losses:

$$\mathcal{P}_{\text{pump}} = k_{\text{pump}} \left(\tfrac{3}{2} n_e T_e + \tfrac{3}{2} n_+^{\text{total}}\, T_i\right) \tag{31}$$

6.6 Surface Coverage Balance

$$\frac{d\theta_X}{dt} = \frac{(\text{production flux} - \text{loss flux}) \cdot V}{A \cdot n_{\text{sites}}} \tag{32}$$

where $n_{\text{sites}} = 10^{19}\,\text{m}^{-2}$ is the surface site density.

7 Transport Equations

7.1 Neutral Wall Loss — Chantry Formula

For a neutral species with total effective sticking probability s_{eff}:

$$\boxed{\nu_{\text{wall}} = \left[\frac{1}{\nu_{\text{surf}}} + \frac{1}{\nu_{\text{diff}}}\right]^{-1}} \tag{33}$$

where:

$$\nu_{\text{surf}} = \frac{s_{\text{eff}}}{2 - s_{\text{eff}}} \cdot \frac{v_{\text{th}}}{4} \cdot \frac{A}{V} \qquad \text{(surface-limited rate)} \tag{34}$$

$$\nu_{\text{diff}} = \frac{D}{\Lambda^2} \qquad \text{(diffusion-limited rate)} \tag{35}$$

with thermal velocity $v_{\rm th} = \sqrt{8k_B T_{\rm gas}/(\pi M)}$, free diffusion coefficient $D = (\pi/8)\,\lambda\, v_{\rm th}$, and mean free path $\lambda = 1/(n_{\rm gas}\,\sigma_{\rm coll})$.

Competing surface reactions: When a species participates in multiple surface reactions, compute:

$$s_{\rm eff} = \sum_j s_j\,\theta_j, \qquad \text{Rate}_j = \nu_{\rm wall}\cdot\frac{s_j\,\theta_j}{s_{\rm eff}}\cdot n_i \tag{36}$$

7.2 Ion Wall Loss — Lee & Lieberman h-Factors

The ion wall loss frequency for species i is:

$$\nu_{i^+}^{\rm wall} = u_{B,i}\cdot\frac{h_L\,A_{\rm ax} + h_R\,A_{\rm rad}}{V} \tag{37}$$

where $u_{B,i} = \sqrt{eT_e/M_i}$ is the Bohm velocity.

Axial h-factor (Lee & Lieberman 1995, Eq. A9):

$$\boxed{h_L = \frac{1+3\alpha_{\rm avg}/\gamma}{1+\alpha_{\rm avg}}\cdot\frac{0.86}{\sqrt{3+\dfrac{L}{2\lambda_i}+\left(\dfrac{0.86\,L\,u_B}{\pi D_a}\right)^2}}} \tag{38}$$

Radial h-factor (Eq. A10):

$$\boxed{h_R = \frac{1+3\alpha_{\rm avg}/\gamma}{1+\alpha_{\rm avg}}\cdot\frac{0.80}{\sqrt{4+\dfrac{R}{\lambda_i}+\left(\dfrac{0.80\,R\,u_B}{2.405\,J_1(2.405)\,D_a}\right)^2}}} \tag{39}$$

where:

$$\alpha_{\rm avg} = n^-/n_e \qquad \text{(electronegativity ratio)} \tag{40}$$

$$\gamma = T_e/T_i \qquad \text{(temperature ratio)} \tag{41}$$

$$\lambda_i = 1/(n_{\rm gas}\,\sigma_{\rm ion}) \qquad \text{(ion mean free path)} \tag{42}$$

$$D_a = eT_e/(M_{\rm ion}\,\nu_i) \qquad \text{(ambipolar diffusion coefficient)} \tag{43}$$

$$J_1(2.405) = 0.5191 \qquad \text{(Bessel function value)} \tag{44}$$

Physical meaning:

- The prefactor $(1+3\alpha/\gamma)/(1+\alpha)$ reduces ion wall flux in electronegative plasmas. At $\alpha \gg 1$, it approaches $3/\gamma \ll 1$, reflecting negative ion confinement.
- The term $3 + L/(2\lambda_i)$ interpolates between free-streaming and collisional transport.
- The term $(0.86\,L\,u_B/\pi D_a)^2$ adds high-pressure ambipolar diffusion.

7.3 Ion Temperature

From Lee & Lieberman (1995):

$$T_i\,[\text{eV}] = \begin{cases} \dfrac{0.5 - T_{\rm gas}/11605}{p\,[\text{mTorr}]} + \dfrac{T_{\rm gas}}{11605} & \text{if } p > 1\,\text{mTorr} \\ 0.5 & \text{otherwise} \end{cases} \tag{45}$$

At operating conditions ($p = 6.9\,\text{mTorr}$, $T_{\rm gas} = 315\,\text{K}$): $T_i \approx 0.096\,\text{eV} \approx 1110\,\text{K}$.

7.4 Negative Ion Transport

Negative ions are confined by the sheath potential barrier and **do not reach the walls**. Their only losses are volume processes: detachment (G20–G21) and ion–ion recombination (G42–G49).

8 Operating Conditions

Parameter	Value
SF_6 feed	100 sccm
Ar feed	10 sccm (actinometer only)
Gas temperature	315 K
Power range	100–3500 W
p_{OFF} range	0.5–4.0 Pa
Pressure controller	OFF (constant outlet flow)
Power coupling efficiency	0.50

Pressure after ignition:

$$p = \left(\sum_{i=1}^{N_{\mathrm{neutral}}} n_i + n_{\mathrm{Ar}} \right) k_B T_{\mathrm{gas}}, \qquad \Delta p = p - p_{\mathrm{OFF}} \tag{46}$$

9 Solver Method

The system of 18 ODEs is integrated using `scipy.integrate.solve_ivp` with the BDF method.

State vector (18 components):

$$\mathbf{y} = \left[\underbrace{n_{\mathrm{SF_6}}, \ldots, n_{e^-}}_{13},\ T_e,\ \underbrace{\theta_{\mathrm{F}}, \ldots, \theta_{\mathrm{SF_5}}}_{4} \right] \tag{47}$$

Initial conditions: Pure SF_6 at p_{OFF}, with trace ions ($10^{10}\,\mathrm{m}^{-3}$), $T_e = 5.0\,\mathrm{eV}$, all $\theta = 0.01$.
Integration time: 1.0 s (sufficient for steady state).
Tolerances: rtol $= 10^{-8}$, atol $= 10^{-6}$, max step $= 10^{-3}$ s.

10 References

1. Kokkoris et al., *J. Phys. D: Appl. Phys.* **42**, 055209 (2009) — Target paper
2. Kokkoris et al., *J. Phys. D: Appl. Phys.* **41**, 195211 (2008) — Power balance equations
3. Lee & Lieberman, *J. Vac. Sci. Technol. A* **13**, 368 (1995) — h-factors
4. Ryan & Plumb, *Plasma Chem. Plasma Process.* **10**, 207 (1990) — Troe fall-off formula
5. Ryan & Plumb, *Plasma Chem. Plasma Process.* **8**, 281 (1988) — Experimental rates
6. Chantry, *J. Appl. Phys.* **62**, 1141 (1987) — Neutral wall loss formula
7. Christophorou & Olthoff, *J. Phys. Chem. Ref. Data* **29**, 267 (2000) — SF_6 cross sections